Highlights

Inattention raises energy prices

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- Rational inattention is embedded in the auctioneer of a two-stage cap-and-trade model.
- Three specifications of the inattentive planner are solved as mixed complementarity problems.
- An inattentive planner issues fewer allowances than the carbon budget authorises.
- The cost of accuracy is passed through to allowance and electricity prices.

Inattention raises energy prices

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ABSTRACT

In a capacity investment framework, we study the regulator's preferences with respect to environmental awareness. Preferences are represented through a structure with rational inattention, following the seminal contribution of Sims (2003): the planner holds a noisy estimate of the emissions CAP and invests in information to reduce the gap between the environmentally sustainable CAP and the actual one. We embed this planner in the two-stage cap-and-trade equilibrium model of Amigo, Cea-Echenique and Feijoo (2021), in which energy generators differ in technology. The planner optimally decides (i) the level of accuracy with respect to the carbon budget and (ii) the initial price of allowances, while generators choose (i) capacity expansion, (ii) production plans and (iii) an allowance trading strategy. We formulate three specifications of the inattentive planner — one profit oriented and two welfare oriented — transform each equilibrium into a mixed complementarity problem and solve them with the PATH solver, calibrated on the Chilean power system. Our main result is that a lack of accuracy may involve higher allowance and electricity prices than situations in which the regulator is less attentive: over the ten carbon budgets we examine, the inattentive planner issues fewer allowances than the budget authorises for the larger budgets and sustains allowance prices between 77% and 88% of those of the original model. Attention is therefore not a free good for the regulator, and its cost is passed on to electricity consumers.

1. Introduction

Pollution control has been a focal point in the past 20 years, with increased awareness and demand for better systems in the last few years. Power generation is a natural target for that effort. In Chile, the country we calibrate our model on, the long-term climate strategy published by the Ministry of the Environment reports that the energy sector accounted for 77% of the greenhouse gases emitted in 2018, and that within that sector electricity generation is the largest single contributor with 30% of total national emissions (Gobierno de Chile, 2021). Any instrument that changes the incentives of electricity generators therefore operates on a large share of a country's emissions.

This sector is commonly regulated by each government by applying a carbon tax and/or cap-and-trade mechanisms. Finland, in 1990, was the first country to implement a carbon tax, since then 19 other countries in Europe and many across the world have done the same according.[1] The tax is charged for every unit of CO_2e and the amount varies for each case.

Alternatively, there is the Cap-and-Trade System (CaTS), concept created by Thomas Crocker in 1966. A somewhat contradictory origin, because in 2009 he told to the Wall Street Journal that he thinks carbon tax is a more effective mechanism because of its easier regulation.[2] Nevertheless, this dichotomy stands on an active literature trying to understands different structures to assess climate policy.

The CaTS is a structure in which a regulatory entity determines a carbon budget from which contamination permits are emitted and consequentially bought or assigned to the polluters (Hanemann, 2010). The amount bought is their limit of contamination.

Even though this two systems may be independent from each other, now a days Europe has a mixed one called Emissions Trading System (EU ETS), including carbon tax and CaTS.[3]

2. Literature review

Studies regarding CaTS effectiveness has been done. Chen, Chen and Ma (2021) proves that applying a CaTS is much more effective in reducing carbon emissions and encourage renewable energy investment and transition in comparison to not implement any cap and trade mechanism. They particularly study no CaTS, Grandfathering rule and Benchmarking rule. They conclude that comparing between Grandfathering and Benchmarking, the first produces less carbon emissions while the second one produced more investment in renewable energy.

Yang, Pan, Ma, Yang and Ke (2020) arrives to the same conclusion. Additionally, they propose "mathematical models to quantify the optimal green technology investment and product pricing, and compare the effects of the two allocation rules on these operational decisions and total emission", resulting in that Grandfathering rule mechanism produces higher optimal product price in comparison to a Benchmarking mechanism price.

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[1]See https://taxfoundation.org/carbon-taxes-in-europe-2022/
[2]Se for instance the article in Business Insider (link)
[3]See https://climate.ec.europa.eu/eu-action/eu-emissions-trading-system-eu-ets_es

These studies show that the importance is in how CaTS is implemented and regulated, because there is clear benefits in its theoretical application.

Three contributions are closer to the mechanism we study. Foramitti, Savin and van den Bergh (2021) compare cap-and-trade against a carbon tax in an agent-based goods market and show that the ranking of the two instruments depends on how agents form expectations rather than on the instruments themselves, so the behaviour attributed to the regulated firms is doing much of the work. Chen, Chen and Ma (2022) study competition and cooperation among firms under a cap-and-trade regulation and find that the allocation rule shapes not only emissions but also the strategic interaction between the regulated firms. Pasten and Schoenle (2016) move the informational friction to the policy maker itself: in a model where attention is scarce, the policy chosen under limited information processing capacity differs systematically from the full-information policy, and the welfare loss is attributable to the informational constraint and not to the instrument. Our contribution belongs to this last line, transposed to a capacity investment equilibrium: we keep the instrument fixed and let the regulator's accuracy be the object of choice.

While Foramitti et al. (2021) study different models for a goods market, Amigo et al. (2021) studies a CaTS implementation in Chile for the electric generation industry. They propose a equilibrium model that involves the producers and a regulator (allowance permit emitter), bringing results for every year from 2018 to 2050 about permits prices, electricity prices, permits distribution among polluters, production quantities and capacity expansion decisions per agent.

We focus on the policy followed by the regulator that Amigo et al. (2021) define by a chance constraint. This paper extend that configuration, in which we look for a new representation of the auctioneer preferences accounting for costly information. In particular, we establish a link with the literature on rational inattention where the decision maker has a limited capacity to process information. This framework provides a mechanism to understand the optimal level of accuracy regarding environmental targets. Our motivation relies on climate responsibility of regulation and the implicit cost produced by lack of climate awareness.

The literature on behavioural preferences of the planner is comparatively thin. Most cap-and-trade models treat the regulator as either a mechanical constraint — issuing exactly the budget it is given — or as a fully informed welfare maximiser. In Amigo et al. (2021) the auctioneer solves

$$\min_{\theta \geq 0} \; -\theta\pi^a + F(\theta) \quad \text{s.t.} \quad \varphi^{-1}(\varepsilon)\sigma + \mu - \theta \geq 0, \tag{1}$$

where the amount of allowances $\theta \in \mathbb{R}_+$ emitted in the first stage is the only decision variable, $F(\theta)$ is the social cost of carbon (Feijoo and Das, 2014), φ^{-1} is the inverse cumulative distribution function of the standard normal, and μ and σ are the mean and standard deviation of the normal distribution from which the carbon budget is drawn. The chance constraint binds at the optimum, so θ collapses onto the budget the regulator was handed and the regulator has no genuine decision left to make. Replacing that constraint by an explicit cost of accuracy is precisely what gives the planner a problem of its own.

The main conclusion from the papers that studied the cap and trade mechanism is that its regulation and effective implementation is key to a positive impact full result in regards to minimizing carbon and increasing green technologies alternatives.

Gabaix (2019) describes that it is "empirically desirable and theoretically doable" to change the classical assumption in rational economics that people process all the information freely available when making a decision. He says that it is not a real situation and that economic models can include the proportion of information considered and try to maximize it in a optimization problem with the idea to better interpret real life in their models.

There is information or signals of a investment or decision making that is noisy, it comes with interference of bad or confusing information that affects the end result. But, there can be applied a concept in the model that cleans that signal by investing in costly information.

Verrecchia (1982) was one of the first to include this concept in a Bayesian model, in which he shows that "there exists a rational expectation competitive equilibrium in which the amount of costly diverse information each trader acquires is endogenous determined." (in a traders market where they can pay for more precise signals).

These studies show the positive impact of investing in information to make better decisions, but the form and model that this information cost and information acquisition takes in reality is not clear. Dewan and Neligh (2020) conducts experiments with simple information acquisition models that at the end showed that under some assumptions on the cost, "gross payoffs to decision makers are non-decreasing". This models are very important for this paper because a version of them is implemented in the Autioneers model to manage it's attention on carbon emissions made by his permit decision making.

3. The capacity investment model

We take the setting introduced by Amigo et al. (2021) with N producers maximizing profits over a finite time horizon. There is uncertainty with respect to the realization of the state of nature in the second period. Thus, producers choose quantities to generate, allowances of carbon emissions and additional capacity to invest in a first stage. The additional capacity becomes available depending on the technology and in a second stage the producers are able to choose new generation levels and re-trade allowances.

The market of allowances is regulated. There is a planner (also referred as auctioneer in this paper) who is taking an optimal decision regarding the allowances availability in the first stage. Given the optimal provision, there is no more provision of allowances in the second stage but a re-trading market is available for each producer to be a part of.

Carbon budget is represented by the maximum level of CO_2e emission established for a period of time. We assume that the true level is unknown and denoted by $CAP \in \mathbb{R}_+$. The planner chooses the amount of available allowances $\theta \in \mathbb{R}_+$ for the producers to purchase at a price π^a for each unit. The price π^a represents the allowances price in stage 1. Since CAP is unknown, the amount of allowances emitted by the planner can surpass the emissions levels of CAP if its estimation is not adequate.

3.1. Timing, demand and notation

Time is indexed by $T_0 := \{0\} \cup T$ with $T := \{1, \dots, \bar{t}\}$, and Ω is a finite set of states of nature with physical probabilities $(Pr(\omega))_{\omega\in\Omega}$ in the simplex

$$\Delta := \left\{ (Pr(\omega))_{\omega\in\Omega} \in [0,1]^{|\Omega|} : \textstyle\sum_{\omega\in\Omega} Pr(\omega) = 1 \right\}.$$

Throughout we write

$$\mathcal{Z} := \mathbb{R}_+ \times \mathbb{R}_+^{T\times\Omega}$$

for the space of a first-stage decision together with its state-contingent continuation.

In the first stage $t = 0$ demand is deterministic and equal to $D(0) \in \mathbb{R}_+$. In the second stage, demand is state and time contingent, $D(t,\omega) \in \mathbb{R}_+$ for $(t,\omega) \in T \times \Omega$, so that the demand profile is $D = \big(D(0), (D(t,\omega))_{(t,\omega)}\big) \in \mathcal{Z}$. The five demand scenarios we use are those of Amigo et al. (2021) and are described in Appendix A.4. Allowances are traded in two markets: θ allowances are auctioned in $t = 0$ at a price π^a, and a secondary market opens once the state of nature is revealed, in which producer i takes a long position $P_i(\omega)$ and a short position $V_i(\omega)$ at a price $\pi^v(\omega)$.

3.2. Producers

There are N producers indexed by i, each identified with a generation technology. Over a horizon of $\bar{t}$ years, the choice set of producer i consists of

- a capacity expansion plan $x_i \in \mathcal{Z}$, with components $x_i(0)$ and $x_i(t,\omega)$;
- a production plan $Q_i \in \mathcal{Z}$, with components $Q_i(0)$ and $Q_i(t,\omega)$;
- allowances $A_i \in \mathbb{R}_+$ bought from the auctioneer in $t = 0$;
- allowances $P_i(\omega) \in \mathbb{R}_+^{\Omega}$ bought and $V_i(\omega) \in \mathbb{R}_+^{\Omega}$ sold in the secondary market.

Electricity prices are $\pi^d := \big(\pi^d(0), (\pi^d(t,\omega))_{(t,\omega)}\big) \in \mathcal{Z}$ and, given parameters $(a_i, b_i) \in \mathbb{R}_+^2$, the revenue function of producer i is $f_i(p,q) = \left(a_i q + \frac{b_i}{2} q^2\right) - p\,q$. Write

$$\mathbb{X} := \mathcal{Z}^2 \times \mathbb{R}_+ \times \mathbb{R}_+^{\Omega} \times \mathbb{R}_+^{\Omega}, \tag{2}$$

$$\Pi := \mathcal{Z} \times \mathbb{R}_+ \times \mathbb{R}_+^{\Omega}, \tag{3}$$

$$\Xi := \mathbb{R}_+^2 \times \mathbb{R}_+^7, \tag{4}$$

for the choice set, the price space and the parameter space of a producer respectively. Producer i chooses $(x_i, Q_i, A_i, P_i, V_i) \in \mathbb{X}$, taking as given prices $(\pi^d, \pi^a, \pi^v) \in \Pi$, the number of hours in a year $\tau \in \mathbb{R}_+$, the discount rate R, the probabilities $(Pr(\omega))_{\omega\in\Omega} \in \Delta$ and its own parameter vector

$$\xi_i := \big((a_i, b_i), I_i, TC_i, TCR_i, CF_i, \bar{Q}_i, RP_i, \varepsilon_i\big) \in \Xi,$$

so as to solve

$$\begin{aligned}
\min_{\mathbb{X}}\ & f_i\big(\pi^d(0), Q_i(0)\big) + A_i\pi^a + I_i x_i(0) \\
&+ \sum_{\omega} Pr(\omega)\Bigg[\pi^v(\omega)\big(P_i(\omega) - V_i(\omega)\big) \\
&+ \sum_{t>0} \frac{TC_i(t)\, f_i\big(\pi^d(t,\omega), Q_i(t,\omega)\big)}{(1+R)^t} \\
&+ \sum_{t>0} \frac{TCR_i(t)\, I_i\, x_i(t,\omega)}{(1+R)^t}\Bigg]
\end{aligned} \tag{5}$$

subject to, for every i, ω and $t > 0$,

$$(CF_i\tau)\Bigg[\bar{Q}_i + x_i(0) + \sum_{t'<t-lag_i} x_i(t',\omega)\Bigg] - Q_i(t,\omega) \geq 0 \quad (\alpha_{i,\omega,t}) \tag{6}$$

$$(CF_i\tau)\bar{Q}_i - Q_i(0) \geq 0 \quad (\kappa_i) \tag{7}$$

$$RP_i - \bar{Q}_i - x_i(0) - \sum_{t>0} x_i(t,\omega) \geq 0 \quad (\psi_{i,\omega}) \tag{8}$$

$$A_i - V_i(\omega) \geq 0 \quad (\beta_{i,\omega}) \tag{9}$$

$$A_i + P_i(\omega) - V_i(\omega) - \varepsilon_i\Bigg[Q_i(0) + \sum_{t>0} Q_i(t,\omega)\Bigg] \geq 0 \quad (\gamma_{i,\omega}) \tag{10}$$

The first two constraints bound generation by installed plus commissioned capacity, where CF_i is the capacity factor and lag_i the number of years a plant of technology i needs to become fully operational. Constraint (8) caps cumulative capacity by the resource potential RP_i. Constraint (9) forbids selling allowances that were not purchased in the first stage, and constraint (10) is the compliance condition: allowances held after re-trading must cover realised emissions at the emission factor ε_i. The Greek letters in parentheses are the dual variables associated with each constraint; they are what links the producers' problems to the auctioneer's problem once the system is written as a complementarity problem.

3.3. The auctioneer

The auctioneer chooses the quantity of allowances $\theta \in \mathbb{R}_+$ made available in $t = 0$ at a price π^a, which is set so that the market for CO_2 permits clears between generators and the auctioneer. The carbon budget $CAP \in \mathbb{R}_+$ — the maximum level of CO_2e emissions allowed in the second stage — is not observed, and is drawn from a normal distribution, $CAP \sim N(\mu, \sigma^2)$. In Amigo et al. (2021) the auctioneer accommodates that uncertainty through the chance constraint $Pr(\theta \geq CAP) \leq \epsilon$, which yields problem (1), reproduced here with its dual variables:

$$\min_{\theta} \; -\theta\pi^a + \mathcal{F}(\theta) \tag{11}$$

$$\text{s.t.} \quad \varphi^{-1}(\epsilon)\sigma + \mu - \theta \geq 0 \qquad (\eta) \tag{12}$$

$$\theta \geq 0 \qquad (\varrho) \tag{13}$$

Since the chance constraint binds, this auctioneer issues the whole budget it is allowed to issue and plays no substantive role in the equilibrium. The value of the allowances directly influences the price of electricity supplied to the public, so how many allowances are issued matters a great deal; replacing the chance constraint by an explicit, costly notion of accuracy is what turns the planner into a decision maker.

3.4. Equilibrium

The data that parameterise the capacity investment model is the tuple

$$\Theta := \Big(\tau, (Pr(\omega))_{\omega\in\Omega}, (CAP, \mu, \sigma, \epsilon), \left(\xi_i\right)_{i\in N}\Big) \in \mathbb{R}_+ \times \Delta \times \mathbb{R}^4_+ \times \Xi^N. \tag{14}$$

Definition 1. An equilibrium in the capacity investment model is a vector of prices and decisions

$$\left((\pi^{d*}, \pi^{a*}, \pi^{v*}), (x_i^*, Q_i^*, A_i^*, P_i^*, V_i^*)_{i\leq N}, \theta^*\right) \in \Pi \times \mathbb{X}^N \times \mathbb{R}_+$$

such that (i) $(x_i^*, Q_i^*, A_i^*, P_i^*, V_i^*)$ solves the problem of producer i for every $i \in \{1, \ldots, N\}$, (ii) θ^* solves the auctioneer's problem, and (iii) the following market clearing conditions hold, with the associated price in parentheses:

$$\sum_i A_i^* = \theta^* \qquad (\pi^{a*}) \tag{15}$$

$$\sum_i P_i^*(\omega) = \sum_i V_i^*(\omega) \quad \forall\,\omega \qquad (\pi^{v*}(\omega)) \tag{16}$$

$$\sum_i Q_i^*(0) = D(0) \qquad (\pi^{d*}(0)) \tag{17}$$

$$\sum_i Q_i^*(t,\omega) = D(t,\omega) \quad \forall\,\omega, t \qquad (\pi^{d*}(t,\omega)) \tag{18}$$

Conditions (15) and (16) clear the primary and the secondary allowance markets; (17) and (18) clear electricity demand in each stage. Each agent's problem is convex, so replacing it by its Karush-Kuhn-Tucker conditions is without loss, and the equilibrium of Definition 1 can be written as a single mixed complementarity problem (MCP) stacking the producers' conditions, the auctioneer's conditions and the market clearing conditions (Ferris and Munson, 2000). We solve every specification below with the PATH solver (Dirkse and Ferris, 1995) in GAMS.

4. Rational inattention in the auctioneer's problem

Following Dewan and Neligh (2020) we let the planner buy accuracy at a quadratic cost,

$$C(P) = \begin{cases} 0, & P \leq d \\ c(P-d)^2, & P > d \end{cases} \tag{19}$$

where $P \in [0,1]$ is a performance metric measuring how closely the issued allowances track the true carbon budget, c is the marginal cost of accuracy and d is the performance the planner attains with freely available public information. Dropping the chance constraint of problem (11) in favour of (19) yields two families of planner. A *profit oriented* planner maximises its own revenue net of the cost of accuracy,

$$\min_{(\theta,P)\in[0,(1+\epsilon)CAP]\times[0,1]} -\theta\pi^a P + c(P-d)^2, \tag{20}$$

whereas a *welfare oriented* planner is penalised for the distance between the allowances it issues and the budget,

$$\min_{\theta\in[0,CAP]} -\theta\pi^a + c\left(1 - \frac{|\theta - CAP|}{CAP} - d\right)^2. \tag{21}$$

The absolute value in (21) is not differentiable, so we implement it in two ways — a quadratic rate and an auxiliary precision variable — which gives the three specifications we take to the solver. Subsections 4.1 to 4.3 state each one; Section 5 reports the results.

4.1. Profit oriented specification

The Profit Oriented version of the auctioneer model is modeled as follows:

$$\min_{\theta} \quad -\theta\pi^a P + c(P-d)^2 + F(\theta) \qquad \text{s.a.} \tag{22}$$

$$\theta\pi^a P - c(P-d)^2 \geq 0 \quad (\eta) \tag{23}$$

$$\hat{\theta}\cdot(1 + error) - \theta \geq 0 \quad (\mu) \tag{24}$$

$$\theta \geq 0 \quad (\varrho) \tag{25}$$

where $\hat{\theta}$ is the government established carbon budget and *error* is the error associated with its calculation. P is the performance of the decision maker, which is included in the model as a parameter to have a exogenous effect on the model. This allows us to understand the effect of the inclusion of the rational attention concept to the model in a more manageable way.

4.2. Quadratic rate welfare oriented specification

The Welfare Oriented version of the auctioneer model absolute value problem was solved in two different ways. The first one is modeled as follows:

$$\min_{\theta} \quad -\theta\pi^a + c(1 - \frac{(\theta-\hat{\theta})^2}{\hat{\theta}^2} - d)^2 \\ \text{s.a.} \tag{26}$$

$$\theta\pi^a - c(1 - \frac{(\theta - \hat{\theta})^2}{\hat{\theta}^2} - d)^2 \geq 0 \qquad (\eta) \tag{27}$$

$$1 - \frac{(\theta-\hat{\theta})^2}{\hat{\theta}^2} - d \geq 0 \qquad (\mu) \tag{28}$$

$$\frac{(\theta-\hat{\theta})^2}{\hat{\theta}^2} + d \geq 0 \qquad (\lambda) \tag{29}$$

$$\theta \geq 0 \qquad (\varrho) \tag{30}$$

where the performance P is now a variable formed as follows:

$$P = 1 - (\frac{\theta - \hat{\theta}}{\hat{\theta}})^2 \tag{31}$$

4.3. Precision welfare oriented specification

The second version in which the absolute value problem is solved, is by transforming the numerator into a variable r. The performance is now defined as $P = 1 - \frac{r}{\hat{\theta}}$. The complete model is as follows:

$$\min_{\theta, r} \quad -\theta\pi^a + c(1 - \frac{r}{\hat{\theta}} - d)^2 \\ \text{s.a.} \tag{32}$$

$$\theta\pi^a - c(1 - \frac{r}{\hat{\theta}} - d)^2 \geq 0 \qquad (\eta) \tag{33}$$

$$r - \theta + \hat{\theta} \geq 0 \qquad (\lambda) \tag{34}$$

$$r + \theta - \hat{\theta} \geq 0 \qquad (\delta) \tag{35}$$

$$1 - \frac{r}{\hat{\theta}} - d \geq 0 \qquad (\mu) \tag{36}$$

$$\frac{r}{\hat{\theta}} + d \geq 0 \qquad (\chi) \tag{37}$$

$$\theta \geq 0 \qquad (\varrho) \tag{38}$$

$$r \geq 0 \qquad (\alpha) \tag{39}$$

Table 1
P effect in θ and π^a Profit Oriented Model CAP($\hat{\theta}$)= 100 million.

Perf[P]	θ	$\frac{\theta}{CAP}$	π^a
0.798	120,200,000	1.20	\$270.13
0.8	120,200,000	1.20	\$270.13
0.85	113,654,321	1.14	\$25.75
0.9	113,568,965	1.14	\$93.65
0.95	118,470,479	1.18	\$188.86
0.99	114,916,396	1.15	\$298.11
1	118,332,486	1.18	\$317.24

5. Results

All three specifications are calibrated against the original model of Amigo et al. (2021) on the Chilean power system, with a reference carbon budget of 100 $MtCO_2e$ and the corresponding allowance price $\pi^a = 315.83$. The performance attainable with public information is set to $d = 0.798$, obtained from the $\epsilon = 0.202$ error in national carbon budget accounting reported by Andres, Boden and Higdon (2014). For each specification the marginal cost of accuracy c is then chosen so that the allowance price it reproduces at that budget is as close as possible to 315.83; this yields c = 920,000 million for the profit oriented planner, c = 9,900 million for the quadratic rate planner and c = 4 million for the precision planner. The three specifications are then solved over ten carbon budgets, from 100 to 1,000 $MtCO_2e$.

5.1. Profit oriented planner

Using the KKT conditions, this problem (see A.1), the producer's model and the market clearing conditions are transformed to a MCP, then, the equilibrium model is solved as one MCP with the PATH solver in GAMS. (Ferris and Munson (2000)).

Because the performance (P) is introduced in the model as a parameter, the first evaluation of the model is made by the variation of P. Andres et al. (2014) calculate the error of the calculations of the Chilean carbon budget as 20.2% and because the definition of P where it can't be greater than 1, the model is first evaluated by the sole variation of P and maintaining constant the other parameters (*ceteris paribus*). The results of this exercise are presented in Table 1.

The price of the permits (variable π^a) and the emitted permits (variable θ) optimal values found by the model distribute in a similar pattern. They appear to have an quadratic distribution associated with the change of the performance value, in which the minimum and maximum values of P produces the maximum emitted permits and prices, but the center performances (0.85,0.9) produce local or global minimum values.

This results show that the performance has a direct effect on the model. One way of analyzing this effect is that there is an optimal performance between its possible values $P \in [0.798, 1]$. Also, investing the maximum in information $P = 1$, can produce non optimal solution for the emitted permits.

Table 2
CAP effect on θ y π^a Profit Oriented Model P = 0.8.

$CAP(\hat{\theta})$	θ	$\frac{\theta}{CAP}$	π^a
100M	120,200,000	1.20	$270.13
200M	240,400,000	1.20	$60.57
300M	360,600,000	1.20	$119.47
400M	454,798,037	1.14	$95.83
500M	601,000,000	1.20	$79.31
600M	605,279,899	1.01	$66.57
700M	605,279,899	0.86	$56.37
800M	961,600,000	1.20	$152.50
900M	605,279,899	0.67	$42.26
1,000M	605,279,899	0.61	$35.98

To understand the implementation of the rational inattention concept in this model, the optimal values of θ and π^a can also be analyzed, thanks to the parametrization of P, by making the performance constant and solve the model for different carbon budgets.

As shown in Table 2, the model shows a decreasing relationship $\frac{\theta}{CAP}$ while increasing the parameter of carbon budget. It becomes clear after a carbon budget of $500MtCO_2$, with a possible outlier in $CAP = 800MtCO_2$. The high permit emission by the Auctioneer in the lower half of carbon budgets may be an indication of the short term energy production capacity of Chile. This model is calibrated with the 2018 electricity production distribution (base model Amigo et al. (2021)). Because of this, the model tries to minimize the electricity prices (variable π^d) by emitting more permits in the lower budgets, but, for larger budgets, the model understands that the system doesn't need that amount of permits and minimizes the carbon emission, demonstrating the first proof of social benefit by introducing the inattention concept.

To make a comparison between performances for each carbon budget, the ratio $\frac{\theta}{CAP}$ found in the model is illustrated in Fig. 1. This results show that at low performances ($P \leq 0.85$) the Auctioneer tends to over-produce permits. But for higher performances ($P > 0.85$), the auctioneer tends to always produce less permits than the carbon budget, specially when the budget is bigger.

Two forces act in opposite directions here. Raising the carbon budget relaxes the upper bound $\hat{\theta}(1 + error)$ in constraint (24) and therefore allows more allowances to be issued, which pushes the allowance price down along the demand schedule. At the same time, the revenue term $-\theta\pi^a P$ in objective (22) is multiplied by the performance, so a planner facing a larger budget can afford to buy more accuracy out of the same revenue — and the cost $c(P-d)^2$ of that accuracy has to be recovered from the generators who buy the allowances. The net effect is that the ratio $\frac{\theta}{CAP}$ falls as the budget grows: at large budgets the profit oriented planner stops issuing everything it is entitled to issue, and the allowance price settles at a level between 77% and 88% of the price of the original model at the same budget. Attention, in other words, is paid for by the buyers of allowances.

To understand what is happening about the permit distribution between the different carbon emitters energy generators (carbon, gas and petro-diesel are considered), the model produces the results in Fig. 2 and Fig. 3.

The first stage is when the Auctioneer decides the amount of permits to produce and sells them to the energy companies. Then, in the second stage, demand scenarios are revealed and a permit market is opened. This model in particular shows marginal change in the permit distribution for almost all carbon budgets. Coal base energy producers appear to have the most permits in both scenarios. There is an anomaly present when $CAP = 800MtCO_2$ (962 million permits), it is possibly a solver error.

Other important variable to consider for the evaluation of this model is the electricity prices for the public per carbon budget. Table 3 shows interesting prices in the first stage, there appears to be an inverse effect on prices when the $\frac{\theta}{CAP}$ ratio increases.

Finally, the prices in the second stage can be evaluated for every demand scenario possible. The description of every demand scenario see A.4. Fig. 4 shows that for the high GDP scenario (scenario 4) the prices are higher. On the other hand, there is an anomaly again in the $800MtCO_2$ budget.

5.2. Quadratic rate welfare oriented planner

Like the previous model, this one is transformed into a MCP using the KKT conditions (see A.2) and solved using the PATH solver in GAMS. The first results are given by changing only the carbon budget in the interval $[100MtCO_2, 1000MtCO_2]$. The resumed results of this exercise are showed in Table 4.

The permits emitted by the Auctioneer seems to find a upper limit. The maximum permits created are $698,676,376$ even though the carbon budget is $1,000,000,000$ in that scenario. This shows that the attention concept inclusion enforces the model to produce around 600 million permits. In the bottom half of the budgets (100 to 400 million) emitting far from that enforced amount, appears to have a relation with low performances. On the other hand, the top 3 budgets (800, 900 and 1000 million) produce around 600 million permits, have a smaller $\frac{\theta}{CAP}$, have higher performances and the smallest permit prices.

To understand the performance results through the carbon budgets, this relationship is graphed in Fig. 5. This graph shows that there has to be a limit in the carbon budget between 400 and 500 million tons of CO_2 in which the performance changes and investment in information is elevated. Also there is a noticeable decrease in performance after the 700 million mark.

When compared with the results in Fig. 6, the decrease in ratio ($\frac{\theta}{CAP}$) coincides with the increase in performance showed in Fig. 5. Like the last model, but in a lesser way, this model tends to decrease the amount of permits emitted the bigger the budget is. This is evidence of social benefit improvements because of the minimization of carbon pollution.

Table 3
Electricity prices (π^d) per CAP and $P = 80\%$

CAP	100M	200M	300M	400M	500M	600M	700M	800M	900M	1000M
θ	120M	240M	361M	455M	601M	605M	605M	961M	605M	605M
π^d	\$264.03	\$119.37	\$136.85	\$95.39	\$102.94	\$83.32	\$83.32	\$174.56	\$83.32	\$83.32

Table 4
Quadratic Rate CAP variation results.

CAP	θ	Perf[P]	$\frac{\theta}{CAP}$	π^a
100,000,000	144,944,410	0.798	1.449	\$232.82
200,000,000	289,888,820	0.798	1.449	\$141.56
300,000,000	434,833,230	0.798	1.449	\$103.33
400,000,000	579,777,640	0.798	1.449	\$-
500,000,000	653,754,577	0.905	1.308	\$37.27
600,000,000	619,939,821	0.999	1.033	\$40.43
700,000,000	661,258,561	0.997	0.945	\$36.42
800,000,000	610,913,181	0.944	0.764	\$31.28
900,000,000	609,127,279	0.896	0.677	\$31.21
1,000,000,000	698,676,376	0.909	0.699	\$30.00

In regards to the permits distribution in this model in each scenario, like the previous model, coal base energy producers tends to be the dominant agent in the market. Fig. 7 and 8 graphs both scenarios of permit buying and distribution.

After the initial permits emission and energy generators buys, there is a equilibrium price found of the electricity for the public. Table 5 shows similar prices for the top 6 carbon budgets (400 to 1000 $MtCO_2$), this may be related to the amount of permits emitted (θ) in those budgets, where the Auctioneer sold around $600MtCO_2$ in each one.

In the second stage, the demand scenarios are showed and the produces adapt their capacities. For each scenario and carbon budget, an average electricity was calculated to compare the scenarios in this model. Fig. 9 evidences similar differences between each scenario in comparison with the Profit Oriented Model. Nevertheless, this model spikes prices for all scenarios in the 700 million budget. This may be a indication that this particular model affects negatively social benefit by high electricity prices in some situations.

5.3. Precision welfare oriented planner

Equations 34 and 35 are included so the absolute value transformation is equivalent to this new definition. The main difference with the other modelations is the inclusion of a second variable. This one, as the others, is transformed into a MCP by the KKT conditions (see A.3), coded into GAMS and solved with the PATH solver.

In this model, the results presented in Table 6, show that in this case the performance was unchanged for all the carbon budgets. With a constant value of 0.798, in a look that means that there was no investment in information to minimize the noise. Nevertheless, the $\frac{\theta}{CAP}$ ratio is different for almost every budget. Because of the way the performance was defined in the previous model, in that case there was a lineal relation between performance and the ratio. Also, there is at least intention of reducing the emitted permits, in most cases they are less than the limit 1.202 ratio, indicating that even though the model says that no additional information was bought, there is an intention to reduce carbon emissions.

The lack of clarity in this model shows there is space to improve it. There is conflicted analysis that can be related to bad formulation or lack of definition of the information acquisition.

Fig 10 shows the ratios value for each carbon budget. The series has a apparent downtrend tendency. Again, this may be just a random result based on the high non lineal nature of the problem or, evidenced by the same tendency in the other two models, there is a direct effect on the auctioneer model to minimize pollution. Specially in larger budgets, because, the model understands there is a capacity to produce less permits.

The first and second stage permits distributions results in this model have some interesting contributions. One of the Chilean NDCs is to eliminate coal as energy fuel, so finding a model in which this product is efficiently eliminated would be of great aid. The results of this model in Fig. 11 and Fig. 12 show that coal ends up second in permit acquisition for all carbon budgets. This evidences a possibility for this model and the inclusion of rational attention in which direct and more specific objectives can be targeted and achieved.

Electricity prices in the first stage (shown in Table 7), present a low variance between CAP 300 and 1000. Complementing with the previous analysis made about the permits distribution, this prices may not variate a lot in the second stage and future prices because Fig. 12 shows that for most of the carbon budgets there are less than 100% of the permits in use. So, in this model, there is more electricity made by the renewable energy which in the short and medium term is more expensive.

Finally, Fig. 13 validates the last hypothesis. Showing that for all scenarios, there is no much variance in the prices. Having a mean close to a price of \$105 .

Table 5
Initial electricity prices (π^d) per CAP in Quadratic Rate Model.

CAP[$MtCO_2$]	100	200	300	400	500	600	700	800	900	1000
θ[Millones]	145	290	435	580	654	620	661	611	609	699
π^d	$232.54	$155.49	$123.21	$93.90	$87.48	$90.44	$105.74	$95.05	$93.38	$94.53

Table 6
Precision Model results.

CAP	θ	Perf[P]	$\frac{\theta}{CAP}$	π^a
100,000,000	120,200,000	0.798	1.202	$270.13
200,000,000	240,400,000	0.798	1.202	$155.15
300,000,000	354,508,234	0.798	1.182	$39.07
400,000,000	435,648,945	0.798	1.089	$33.93
500,000,000	590,042,475	0.798	1.180	$38.83
600,000,000	721,200,000	0.798	1.202	$39.20
700,000,000	673,836,382	0.798	0.963	$35.74
800,000,000	757,247,245	0.798	0.947	$35.74
900,000,000	851,903,151	0.798	0.947	$35.74
1,000,000,000	947,173,004	0.798	0.947	$35.74

5.4. Comparison across specifications

For this final part, we make a comparison between the 3 models presented in this work and the results of the original model proposed by Amigo et al. (2021). The idea of this is to understand the impact of including the rational inattention concept into the model by comparing the models profits, social benefits and main results about prices and permits. This comparison can result in conclusions about what is the best model or more suitable for the Chilean NDC and other countries NDCs.

The work made in this paper is a first inclusion of the rational inattention concept introduced in a model of this type, maybe the first made in the electric industry. Because of this, this models are not to be used in real applications but are made for research only. We established as the main objective to identify if the inclusion of rational inattention does enhance social benefit by resulting in lower prices, lower permits emitted and lower carbon emissions. To quantify this, we established as a objective that one the the versions of the models presented in this work has to have less permits and lower prices in 40% of the cases.

In Fig. 14, the results show that the original model produce higher profits for the auctioneer in 70% of the carbon budgets. On the one hand, it is important to mention that no expenses were considered for the auctioneer in this model since its objective function only considered a cost called the Social Cost of Carbon. The former was ignored in the equilibrium model solution since it was considered independent of the permissions issued (θ) by the original authors.

On the other hand, the reformulated models take the value of the costs of additional information c by calibrating them with the original. Therefore, taking into account the auctioneer's profits for these models is also not the most important indicator to consider. Additionally, considering profits as an indicator for success in a public program is generally not the main objective, its overhead ratio may be a better one. In a later work, it is expected to calculate these costs in a real way and consider the operational costs for a real time implementation of the model.

Permit sales prices are an indicator of the quantity of permits issued since these obey the law of demand. They are also an indicator of the costs associated with the permit production. Fig. 15 shows that the original model has the largest prices for all carbon budgets. This may be an indicator that the new models are looking for with a greater effect on social welfare, since, despite the fact that these have production costs, they do not transfer these costs to the permits, potentially producing, lower electricity costs.

On the contrary, these prices may be the product of more permits issued, and therefore, more emissions into the environment, producing more carbon and affecting social wellness.

On the contrary, these prices may be the product of more permits issued, and therefore, more emissions into the environment, producing more carbon and affecting social wellness.

Figure 16 shows that the previous analysis is not true for all cases. In the first 5 established CAPs, the auctioneer in the original model issues fewer permits than the new versions, but, in $CAP = 600$ the Square Rate model issues a very similar amount of permits and in the following ones, it issues fewer permits, resulting in lower emissions. Also, the other two models (Precision and Profit Oriented) issue fewer permits in 3 and 4 of the last 5 CAPs, respectively.

Thanks to the previous statement, the criteria mentioned at the beginning of this subchapter is met to validate the use of rational care in this system, since one of the models has fewer permits issued in more than 40% of the CAPs, but the Most notable is that all models have 3 or more CAPs with lower emissions than the original model.

Finally, electricity prices for individuals are compared between the models in each carbon budget. Figure 17 demonstrates that the Profit Oriented and Quadratic Rate models create lower prices for individuals in 60% of the CAPs. In particular, the Profit Oriented does it for 80% of them.

It is important to note that this last statement is true, partially, because of higher permits emitted than the original model so non green companies create more electricity (its cheaper), in many carbon budgets this is not the case. This brings into perspective that the inclusion of rational inattention does decrease energy prices and impacts positively in social well being.

Table 7
Initial electricity prices (π^d) per CAP in Precision Model.

CAP[$MtCO_2$]	100	200	300	400	500	600	700	800	900	1000
θ[Millones]	120	240	355	436	590	721	674	757	852	947
π^d	\$264.03	\$169.98	\$106.88	\$98.92	\$110.36	\$93.00	\$105.33	\$99.07	\$97.95	\$92.09

It is important now to evaluate why is this happening, why does the model create cheaper electricity prices but at the same time reduces carbon?

6. Conclusion

We have replaced the chance-constrained auctioneer of Amigo et al. (2021) by a planner that must pay for the accuracy of its estimate of the carbon budget, and solved the resulting capacity investment equilibrium as a mixed complementarity problem under three specifications of that cost.

The first result is structural. In the original formulation the chance constraint binds, the auctioneer issues exactly the budget it is handed, and it has no decision to make. Once accuracy is costly, the planner's problem acquires interior content: across the ten carbon budgets we solve, the ratio $\frac{\theta}{CAP}$ declines as the budget grows, and for the largest budgets all three specifications issue substantially fewer allowances than the budget authorises. A planner that has to pay attention emits less than a planner that does not, and it does so without any additional environmental constraint being imposed on it.

The second result concerns prices, and it is the reason for the title. The cost of accuracy does not stay with the planner: it is recovered through the price of allowances and passed on to generators, who pass it on to electricity consumers. Over the ten budgets, allowance prices under the profit oriented planner sit between 77% and 88% of the prices of the original model, and first-stage electricity prices differ across specifications by margins that are economically meaningful. Lack of accuracy is therefore not a neutral feature of the regulator: it shows up in the electricity bill.

The third result is a caveat, and it is where the exercise stops. In the profit oriented specification the performance P is a parameter, not a choice: we solve the equilibrium along a grid of values of P and read off the comparative statics. That specification is thus a sensitivity analysis with respect to attention rather than an implementation of rational inattention in the strict sense. In the precision specification, where P is genuinely endogenous, the solution remains at the corner $P = 0.798$ for every carbon budget, that is, the planner never acquires information beyond what is publicly available. The quadratic rate specification is the only one in which accuracy is both endogenous and interior, and even there the mapping from the budget to performance is not monotone. We read this as evidence that the cost function (19), borrowed from an experimental setting, is not yet the right primitive for a regulator: it prices accuracy but says nothing about the technology that produces it.

That suggests the natural next step. Andres et al. (2014) document that national carbon budget accounting carries a measurable and systematic error, which is the empirical object our performance metric is standing in for. A specification in which the planner invests in reducing *that* error

— with an explicitly modelled information technology, and with the monitoring and enforcement costs a real regulator faces — would make the trade-off between accuracy and welfare identifiable, and would let the model speak to policy rather than only to the mechanism. Extending the planner's role to buying back allowances, or to issuing additional ones in later periods when the state of nature warrants it, points in the same direction.

A. Appendix A

A.1. MCP Auctioneer Profit Oriented

$$0 \leq -P\pi^a(1+\eta) + \mu \perp \theta \geq 0 \tag{40}$$

$$0 \leq \theta\pi^a P - c(P-d)^2 \perp \eta \geq 0 \tag{41}$$

$$0 \leq \hat{\theta}(1+error) - \theta \perp \mu \geq 0 \tag{42}$$

A.2. MCP Auctioneer Welfare Oriented Quadratic Rate

$$0 \leq \mu \perp 1 - \frac{(\theta-\hat{\theta})^2}{\hat{\theta}^2} - d \geq 0 \tag{44}$$

$$0 \leq \lambda \perp \frac{(\theta-\hat{\theta})^2}{\hat{\theta}^2} + d \geq 0 \tag{45}$$

$$0 \leq \eta \perp \theta\pi^a - c(1 - \frac{(\theta-\hat{\theta})^2}{\hat{\theta}^2} - d)^2 \geq 0 \tag{46}$$

A.3. MCP Auctioneer Welfare Oriented Precision Model

$$0 \leq \theta \perp -\pi^a - \eta\pi^a + \lambda - \delta \geq 0 \tag{47}$$

$$0 \leq r \perp 2c\frac{(-1+d+\frac{r}{\hat{\theta}})}{\hat{\theta}} + 2c\eta\frac{(-1+d+\frac{r}{\hat{\theta}})}{\hat{\theta}} - \lambda - \delta \geq 0 \tag{48}$$

$$0 \leq \eta \perp \theta\pi^a - c(1 - \frac{r}{\hat{\theta}} - d)^2 \geq 0 \tag{49}$$

$$0 \leq \lambda \perp r - \theta + \hat{\theta} \geq 0 \tag{50}$$

$$0 \leq \delta \perp r + \theta - \hat{\theta} \geq 0 \tag{51}$$

$$0 \leq \mu \perp 1 - \frac{r}{\hat{\theta}} - d \geq 0 \tag{52}$$

$$0 \leq \chi \perp \frac{r}{\hat{\theta}} + d \geq 0 \tag{53}$$

A.4. Electricity demand scenarios

These are the five electricity demand scenarios defined by Amigo et al. (2021) and reproduced in Fig. 18:

Each scenario is a projection of aggregate electricity demand in Chile between 2015 and 2050. *Low GDP* and *High GDP* bracket the range implied by pessimistic and optimistic growth paths; *Mid GDP* is the central growth projection. *Electrification* adds to the central path the additional demand generated by the substitution of electricity for fossil fuels in transport and heating, which is why it separates from *Mid GDP* after 2030 and approaches the high growth path by 2050. *High effort* is the demand path consistent with an aggressive energy efficiency programme, and is the only scenario in which demand flattens towards the end of the

$$0 \leq \theta \perp -\pi^a + c(-2\theta + 2\hat{\theta})\frac{(2 - 2d - 2(\frac{\theta-\hat{\theta}}{\hat{\theta}})^2)}{\hat{\theta}^2} - \eta\pi^a + c\eta((-2\theta + 2\hat{\theta})\frac{(2 - 2d - 2(\frac{\theta-\hat{\theta}}{\hat{\theta}})^2)}{\hat{\theta}^2}) - \mu(\frac{-2\theta + 2\hat{\theta}}{\hat{\theta}^2}) - \lambda(\frac{2\theta - 2\hat{\theta}}{\hat{\theta}^2}) \geq 0 \tag{43}$$

horizon. Every specification of the planner is solved over all five scenarios, with the probabilities $Pr(\omega)$ of Amigo et al. (2021).

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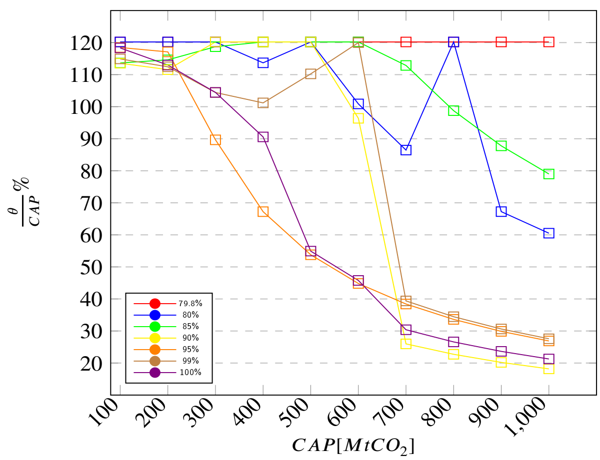

Figure 1: Ratio per CAP in Profit Oriented Model.

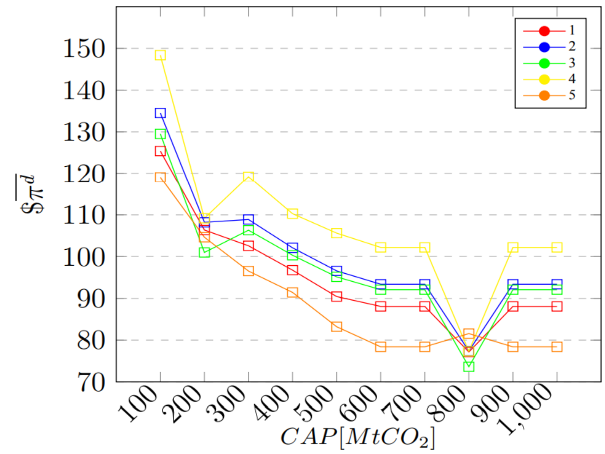

Figure 4: Average electricity prices(2019-2050) per demand scenario

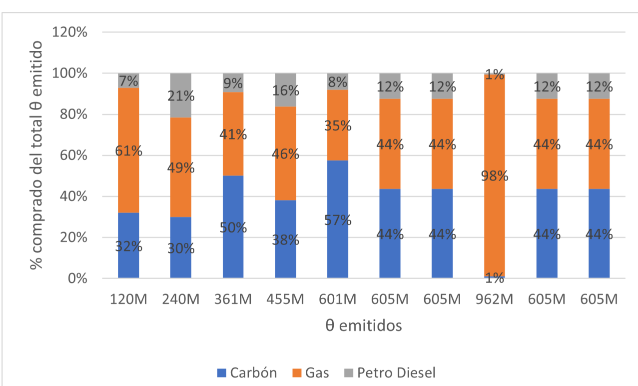

Figure 2: First stage and $P = 80\%$.

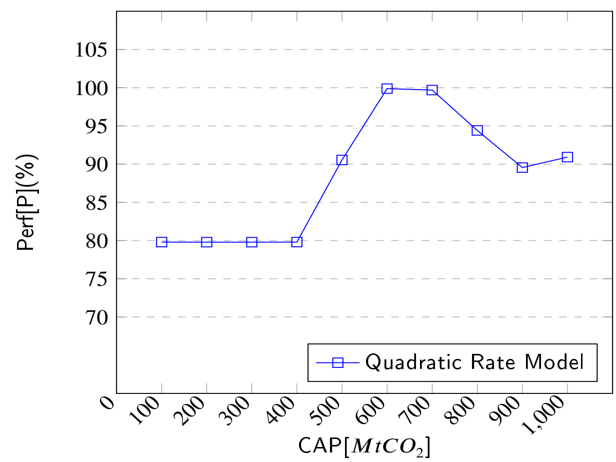

Figure 5: Carbon budget effect on performance.

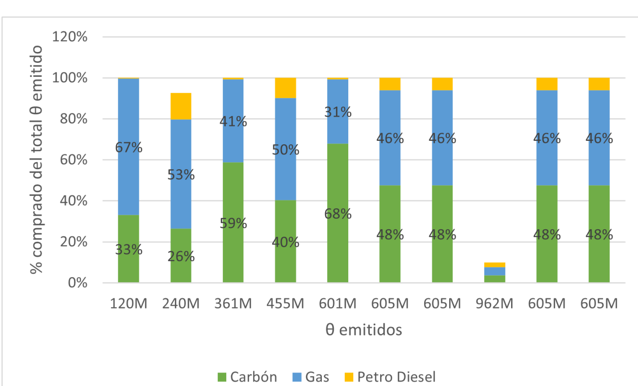

Figure 3: Second stage and $P = 80\%$.

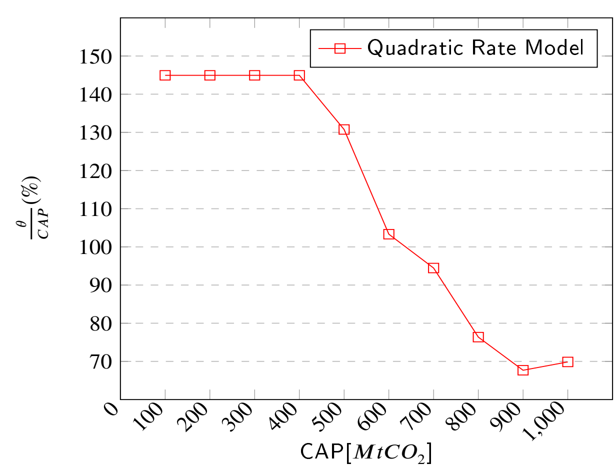

Figure 6: Carbon budget effect on ratio

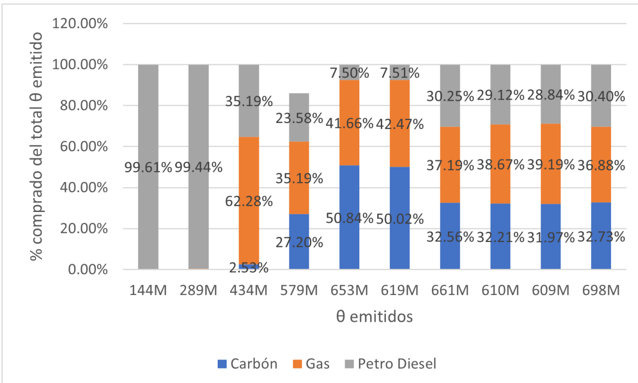

Figure 7: First stage distribution.

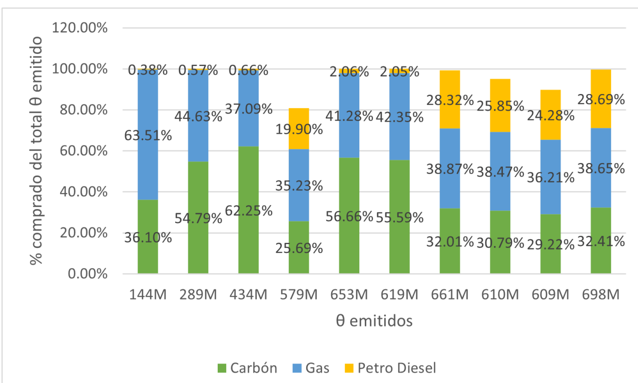

Figure 8: Second stage distribution.

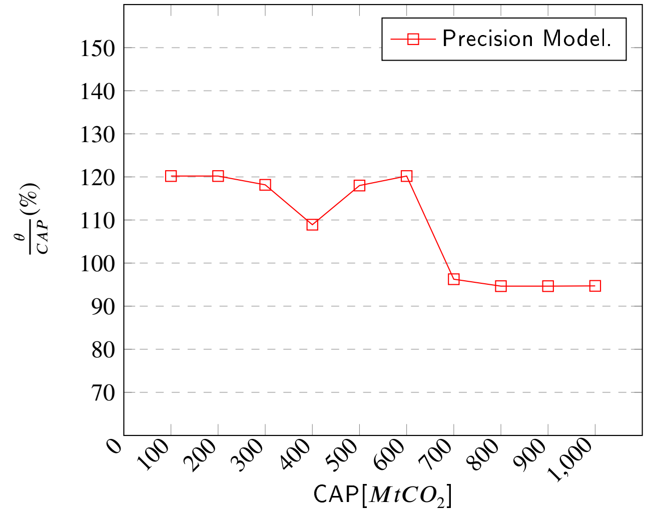

Figure 10: CAP effect on ratio.

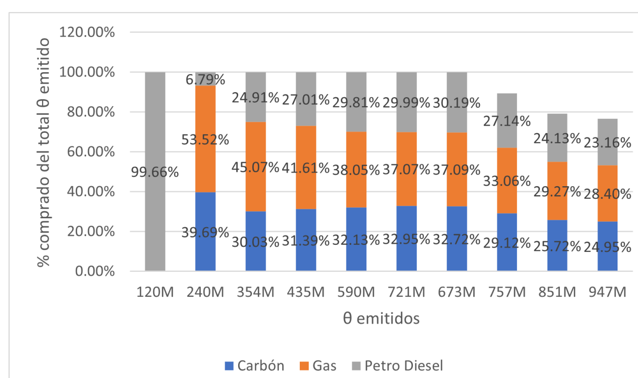

Figure 11: First Stage Distribution.

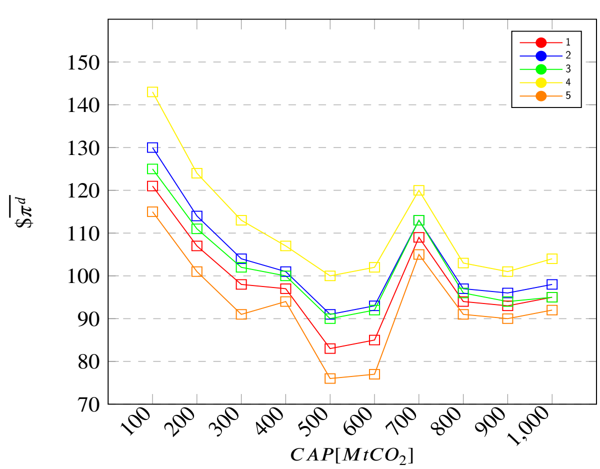

Figure 9: Average electricity prices (2019-2050) per demand escenario Quadratic Rate Model.

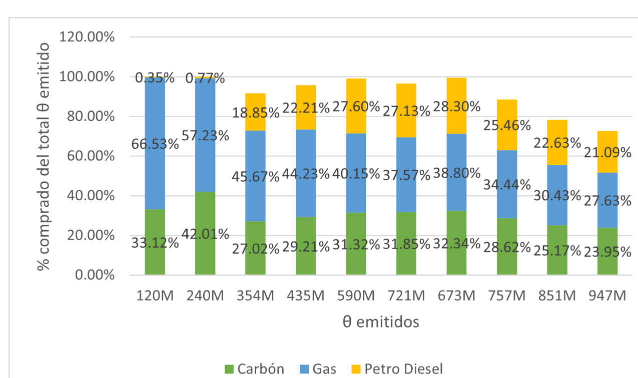

Figure 12: Second Stage Distribution.

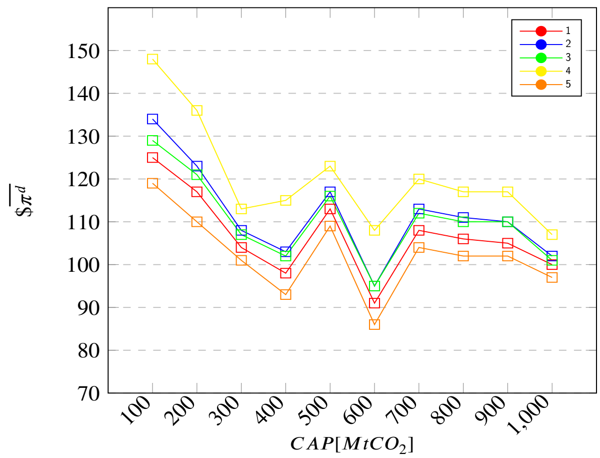

Figure 13: Electricity prices (2019-2050) per scenario

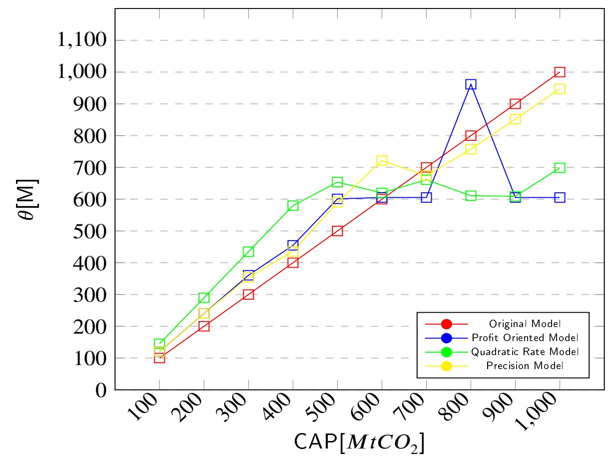

Figure 16: Permits emitted per CAP

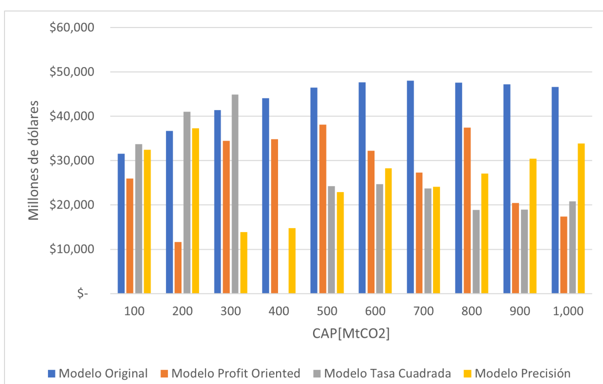

Figure 14: Auctioneer profit per model.

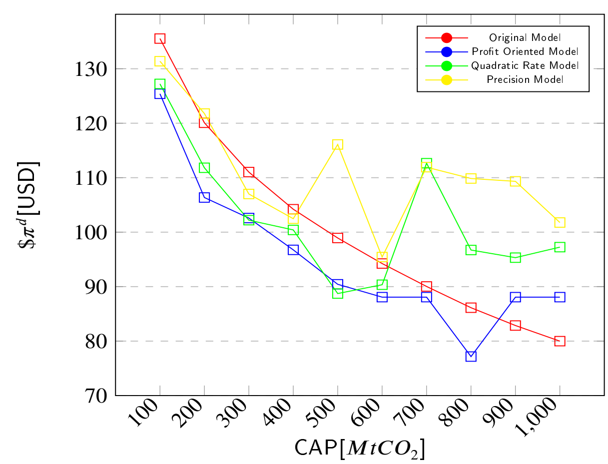

Figure 17: Electricity average price (2019-2050) π^d per CAP.

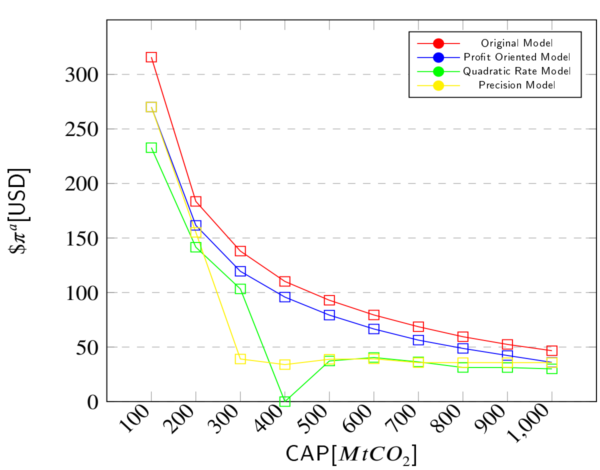

Figure 15: Permit prices π^a per CAP

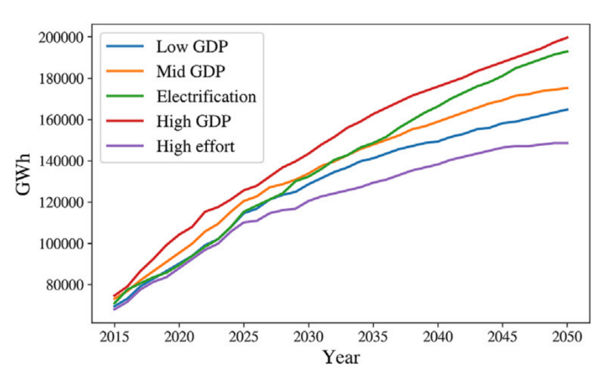

Figure 18: Demand scenarios.